mdstats 0.20.182a0

SIZE-HALVE1 target-size correction

mdstats project

2026-08-15

Summary

`0.20.182a0` corrects target-size selection. Coverage is a hard admissibility limit, not a proxy for training sufficiency. Every coverage-qualified size now enters a 3-epoch target-only learning screen before successive reduction at 10 and 30 epochs.

Scientific authority

The generated funnel is

$$7 \to N_{\text{coverage qualified}} \underset{\text{3 epochs}}{\to} \leq 4 \underset{\text{10 epochs}}{\to} 2 \underset{\text{30 epochs}}{\to} 1.$$

TARGET-DATA2C advances to v3 full-ladder authority. TARGET-DATA2D advances to v2 3/10/30 authority. TARGET-DATA2E advances to v2 full-funnel provenance. Historical PERF-P2/TARGET-DATA2C v2 evidence remains readable as archival qualification but is stale for current generated campaigns.

Epoch-3 screen

The coarse screen uses one fixed leakage-safe target-only EVAL2 role, default 256 configurations, deterministically balanced across development correlation blocks. Replay inference, checkpoint-rescue search, bootstrap selection, and physical verification are not purchased at epoch 3.

Generated preflight requires the 3-epoch endpoint to lie strictly past LR warm-up. With current defaults, $3/30 = 0.10 > 0.05$.

Exact continuation

Survivors continue the same nominal 30-epoch trajectory. Evidence authenticates checkpoint/model, optimizer/scheduler, and Python/NumPy/Torch RNG state at both 3->10 and 10->30 boundaries. Optimizer updates and structures presented are persisted alongside epoch count.

Boundary convergence safeguard

At epoch 3 and 10, the largest hard-coverage boundary is preserved within its practical-equivalence band. It may still be eliminated when materially worse. At epoch 30, smaller-size preference is restored within final practical equivalence. This prevents an early tie from hiding bounded-ladder nonconvergence.

Fidelity calibration requirement

SIZE-FIDELITY1 is now mandatory before PERF-P2R. It must use exhaustive 30-epoch calibration trajectories to verify epoch-3/10 survivor recall across frozen screening seeds and to choose the smallest common coarse monitor that preserves the full-role promotion decision up to practical equivalence. The current 256-frame coarse monitor and 1.0 meV/A coarse equivalence width are provisional defaults, not yet authorizing empirical claims.

Performance roadmap

PERF-P2′s historical lazy-truncation speedup is no longer a current campaign claim. SIZE-FIDELITY1 is inserted before PERF-P2R, and PERF-P2R remains before PERF-P3 to optimize full-ladder coverage and preprocessing reuse, target-only coarse evaluation, exact pause/resume, stage-aware scheduling, checkpoint retention, and complete funnel performance.

No authorizing MACE/GPU runtime was supplied in this environment. GPU training speed or VRAM benefits are therefore not inferred.

Documentation

The canonical architecture advances to revision 48 and dependency-graph schema 30. The normative correction is `docs/specs/training_data/mlff_size_halve1_target_size_revision_spec.md`.